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Subtle Discrimination

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1 Summary

This paper introduces and formalizes the concept of **subtle discrimination**: biased acts that cannot be objectively verified as discriminatory. The key idea is plausible deniability. A decision-maker may favor one group when candidates are close in observable qualifications, but the decision can still be rationalized by unobservable or subjective criteria. This is different from overt discrimination, where a decision is so inconsistent with objective qualifications that bias can be inferred from the decision itself.

The paper develops a promotion model with two ex ante identical agents, called Blue and Red. The principal has a slight bias in favor of Blue. Agents invest in observable human capital before competing for promotion. The bias does not simply lower Red’s promotion chances mechanically. It also changes Red’s and Blue’s investment incentives. Two forces drive the results. The **discouragement effect** reduces investment by the unfavored Red agent when promotion seems unlikely. The **overcompensation effect** makes Red invest more when an extra observable achievement helps her avoid being in a close tie where bias matters.

The main prediction is state-dependent. In **low-stakes** promotion contests, the overcompensation effect dominates: Red invests more than Blue to separate herself from Blue. In **high-stakes** promotion contests, the discouragement effect dominates: Red invests less than Blue because the biased promotion rule makes success too unlikely. This generates different signs of the achievement gap depending on the stakes of the career path.

The model also links discrimination to firm type. Low-productivity firms, or firms for which human capital investment has low returns, may benefit from subtle discrimination because the overcompensation effect pushes discriminated workers to work harder. High-productivity firms lose from subtle discrimination because it discourages discriminated workers from investing in valuable skills. This creates polarization: high-productivity firms choose progressive antidiscrimination policies and have more diversity at the top; low-productivity firms tolerate or even prefer conservative environments and have larger promotion gaps.

The paper uses the framework to interpret several empirical settings: analyst careers, credit decisions, fund flows, banking careers, entrepreneurship, and academia. A broad lesson is that subtle and overt discrimination have different empirical signatures. Outcome tests and average performance comparisons may fail to detect subtle discrimination because subtle bias operates exactly where agents are close substitutes.

2 Big picture

- **Question.** How does discrimination affect human capital investment when biased decisions are plausibly deniable rather than openly discriminatory?
- **Main answer.** Subtle discrimination changes incentives before the decision is made. It can make unfavored agents either overinvest or underinvest depending on promotion stakes.
- **Central distinction.** Overt discrimination is objectively identifiable from the decision; subtle discrimination is hidden behind plausible subjective justifications.
- **Core mechanism.** A biased tie-breaking rule creates both discouragement and overcompensation incentives for the unfavored group.
- **Main firm-level prediction.** High-productivity firms choose progressive policies, while low-productivity firms may tolerate or choose subtle bias.

- **Empirical implication.** Equal or superior performance by discriminated groups does not imply the absence of discrimination; it may reflect overcompensation.

3 Incremental contribution of the paper

- **New taxonomy of discrimination.** The paper classifies discriminatory acts by transparency: subtle versus overt, rather than only by cause, such as taste-based or statistical discrimination.
- **Promotion model with endogenous skill investment.** Discrimination affects not only selection conditional on qualifications, but also the qualifications that candidates choose to acquire.
- **Different predictions for subtle and overt bias.** Overt bias discourages the unfavored group more directly, while subtle bias can induce overcompensation when candidates can avoid close comparisons.
- **Firm polarization result.** The same small bias can be profitable in low-productivity firms but costly in high-productivity firms, generating different corporate cultures and diversity outcomes.
- **Applications across finance.** The framework explains evidence from analysts, lending, fund flows, banking careers, entrepreneurship, and academia.

4 Definitions and intuition

4.1 Decision environment

A decision-maker chooses between two candidates, Blue and Red. Their observable skills are s_b and s_r , and the paper normalizes the skill gap as

$$\Delta s \equiv s_r - s_b \geq 0.$$

The decision-maker also privately observes subjective signals x_b and x_r . Let $\Delta x = x_r - x_b$ and let $F(\cdot)$ be the distribution of Δx . If the decision-maker is unbiased, Blue is chosen only when the subjective signal advantage for Blue is sufficiently large.

4.2 Bias

The paper defines the bias toward Blue as

$$b(s_b, s_r, \omega) = P(s_b, s_r, \omega) - F\left(\frac{s_b - s_r}{\omega}\right) \geq 0,$$

where $P(s_b, s_r, \omega)$ is the probability that the decision-maker chooses Blue, and $F((s_b - s_r)/\omega)$ is the probability that an unbiased decision-maker would choose Blue.

4.3 Subtle versus overt bias

- **Subtle bias.** Bias is subtle when there exist subjective-signal realizations that can justify choosing Blue. In that case, even a biased decision can be rationalized as based on private judgment.

- **Overt bias.** Bias is overt when choosing Blue cannot be justified by any possible subjective signal given the observable skill gap.
- **Plausible deniability.** Subtle discrimination works because a single choice, or a small number of choices, cannot conclusively prove discrimination.
- **Where subtle discrimination is most likely.** It is most relevant when observable qualifications are close, so that subjective assessment can plausibly break the tie.

5 Benchmark: first-best investment

The first-best benchmark is an unbiased social planner who chooses Blue’s and Red’s investments to maximize total surplus:

$$\max_{(e_b, e_r) \in [0,1]^2} \ell + H(e_b + e_r - e_b e_r) - c(e_b) - c(e_r).$$

The term $e_b + e_r - e_b e_r$ captures the probability that at least one agent acquires the relevant skill. There is a trade-off between effort sharing and effort duplication. If both agents invest, marginal effort costs can be shared. But if both agents acquire the same skill, some investment can be duplicative.

The first-best therefore has two possible forms. If effort-sharing benefits dominate duplication costs, the planner asks both agents to invest symmetrically. If duplication costs dominate, the planner asks only one agent to invest. This benchmark is useful because the promotion model later shows that subtle discrimination can push the economy away from both symmetric efficient investment and efficient asymmetric specialization.

6 Promotion model

6.1 Basic structure

The firm must promote one of the two agents. Agents first choose investment levels e_b and e_r . Investment increases the probability of becoming skilled, but it is costly. The principal can reward promotion with a premium W , and the ratio of the promotion premium to investment cost is summarized by

$$\sigma = \frac{W}{k}.$$

The parameter σ is the paper’s measure of **promotion stakes**. High σ means the promotion prize is large relative to the cost of human capital investment.

6.2 Subtle bias in the promotion rule

The principal has a bias $\beta \in [0, 0.5]$ in favor of Blue. The bias matters most when Blue and Red are close in skill. Red can avoid some biased tie-breaking by investing enough to become clearly better than Blue. This is the root of the overcompensation channel.

6.3 Equilibrium investment

In the interior case, equilibrium investment levels take the form

$$e_b^* = \frac{\sigma(0.5 - \beta) + 2\beta\sigma^2(0.5 + \beta)}{1 + 4\beta^2\sigma^2},$$

$$e_r^* = \frac{\sigma(0.5 + \beta) - 2\beta\sigma^2(0.5 - \beta)}{1 + 4\beta^2\sigma^2}.$$

These formulas summarize the central mechanism. A larger σ raises the value of promotion, but the sign of the investment gap depends on whether overcompensation or discouragement dominates.

7 Discouragement versus overcompensation

7.1 Overcompensation effect

When stakes are low, Red can cheaply improve observable qualifications and avoid close comparisons where the principal's bias favors Blue. The marginal benefit of separating from Blue is high relative to the promotion prize. Red invests more than Blue:

$$e_r^* > e_b^*.$$

This is the paper's overcompensation effect. It means that discriminated candidates may appear more qualified than favored candidates before promotion.

7.2 Discouragement effect

When stakes are high, Blue also invests aggressively because the promotion prize is valuable. Red anticipates that even substantial investment may not overcome the biased promotion rule. The expected return on Red's human capital investment falls, and Red invests less than Blue:

$$e_r^* < e_b^*.$$

This is the discouragement effect. It generates the familiar underinvestment channel in high-stakes careers.

7.3 Achievement gap

The achievement gap is the difference in accumulated human capital or visible qualifications. Under subtle discrimination, the sign of the achievement gap changes with stakes. In low-stakes careers, the unfavored group may have higher visible achievement. In high-stakes careers, the favored group has higher visible achievement.

7.4 Promotion gap

The promotion gap can be decomposed into an achievement gap and a favoritism gap. The achievement gap comes from investment choices before promotion. The favoritism gap comes from the biased decision rule conditional on qualifications. Figure 2 shows that the total promotion gap can be nonmonotonic in σ because these components move differently as stakes rise.

8 Firm problem and antidiscrimination policy

8.1 Choosing promotion stakes

The firm chooses the promotion premium-cost ratio σ to maximize profit. In the unbiased case, the firm operates only when the productivity-cost ratio θ is high enough. With subtle bias, the firm trades off stronger worker investment incentives against the cost of paying a higher promotion premium.

8.2 Productivity-cost ratio

The key firm-level parameter is

$$\theta,$$

which measures the productivity benefit of agent skill relative to the cost of inducing investment. High- θ firms are more human-capital intensive and more profitable. Proposition 6 implies that the optimal promotion stake is increasing in θ . Thus, high-productivity firms offer high-stakes career paths.

8.3 Optimal subtle bias

The paper then lets the firm choose its bias β , interpreted as choosing an antidiscrimination policy, supervisory culture, or diversity climate. The result is stark: the optimal bias is a corner solution. Depending on θ , the firm chooses either maximum bias or no bias.

Low- θ firms choose a high subtle bias because the overcompensation effect raises Red's investment in low-stakes settings. High- θ firms choose zero subtle bias because discouragement in high-stakes careers destroys valuable human capital investment by discriminated workers.

8.4 Firm polarization

This creates firm polarization. High-productivity firms prefer progressive antidiscrimination policies and more diversity at the top. Low-productivity firms prefer or tolerate conservative cultures and have less diversity at the top. Small differences in productivity can generate large differences in corporate culture because the optimal bias jumps from one corner to another.

9 Empirical predictions

9.1 Prediction 1: sign of effort gap depends on stakes

The model predicts that discriminated workers invest more than favored workers in low-stakes careers but less in high-stakes careers. Therefore, evidence that the unfavored group has higher pre-promotion performance does not necessarily reject discrimination. It may indicate overcompensation.

9.2 Prediction 2: promotion gaps are larger at the top

The model predicts that high-stakes careers should exhibit stronger discouragement and larger promotion gaps. This helps explain leaky-pipe patterns, where representation gaps grow at higher levels of the hierarchy.

9.3 Prediction 3: high-productivity firms should have stronger diversity policies

High-productivity, high-valuation, financially strong, and human-capital intensive firms have more to lose from discouraging discriminated employees. They should therefore invest more in progressive policies and should have more diverse top management.

9.4 Prediction 4: outcome tests may miss subtle discrimination

Subtle discrimination often operates among close substitutes. If a biased decision-maker selects among candidates or funds that are similar in expected performance, outcome tests may show little performance difference even when discrimination affects allocations.

10 Further applications

10.1 Equity and credit analysts

Analysts are rewarded for both accuracy and optimism. Accuracy is objective, while optimism or client-friendliness is more subjective. If subtle bias affects the subjective dimension, unfavored analysts may overinvest in accuracy. This helps interpret evidence that female analysts can be more accurate than male analysts even while facing career disadvantages.

10.2 Lending

In lending, automated hard-information systems reduce the scope for overt discrimination, but discretion remains when soft information or exceptions are involved. Subtle bias can generate lower approval rates for minority borrowers even when approved minority borrowers do not look safer ex post. The model also predicts that expected discrimination can discourage application effort or shopping for better loan terms.

10.3 Fund flows

Investors may use fund manager characteristics to break ties among similar funds. In such a setting, discrimination can affect flows without producing clear return differences between favored and unfavored managers. This explains why performance-based outcome tests may fail to detect discrimination in fund allocation.

10.4 Banking careers

Gender promotion gaps in banking can arise from subjective assignment, leadership opportunities, and promotion evaluation. The model predicts that gaps should be especially costly in high-stakes segments and that effort responses should differ depending on whether bias is subtle or overt.

10.5 Entrepreneurial finance and academia

For founders, expected bias in later financing stages can affect early human capital and startup investment choices. For academia, subjective evaluation of potential, credit for coauthorship, and grant-review criteria can create subtle discrimination. The model's overcompensation logic helps interpret evidence that women may produce more polished written work or stronger observable signals.

11 Interpretation and intuition

- **Subtle discrimination is not weak discrimination.** The bias can be small at the decision stage but large in its ex ante incentive effects.

- **The key object is not only the promotion decision.** The paper studies how expected bias changes pre-promotion investment.
- **Close calls matter.** Bias is easiest to hide when candidates are similar; this is also where extra observable achievement can help the unfavored group.
- **Performance evidence is ambiguous.** Higher performance by the unfavored group may reflect stronger ability, stronger incentives, or costly overcompensation.
- **Firm culture can be strategic.** Firms may choose to appear progressive or conservative because worker beliefs about bias affect investment.
- **Subtle and overt bias are empirically different.** Treating all discrimination as a single object can obscure the sign of effort and performance gaps.

12 Caveats and discussion

- **Theory model.** The paper provides a conceptual and theoretical framework rather than a direct empirical test of one setting.
- **Binary labels and skills.** The main model uses Blue/Red labels and simplified skill acquisition for tractability, although the paper discusses robustness in the Internet Appendix.
- **Beliefs about bias.** Agents must form beliefs about subtle bias even though individual biased decisions cannot be objectively verified.
- **Policy interpretation.** Quotas or diversity policies may change outcomes, incentives, and welfare differently depending on whether the underlying bias is subtle or overt.
- **Measurement challenge.** Empirical work needs to distinguish achievement gaps caused by ability, discouragement, overcompensation, and direct favoritism.

13 Notes-created summary tables

13.1 Research design at a glance

Element	Implementation	Purpose
Concept	Subtle discrimination: biased acts that cannot be objectively ascertained as discriminatory.	Provides a transparency-based taxonomy of discrimination.
Main setting	Promotion contest between Blue and Red candidates.	Studies how bias affects skill investment before selection.
Key state variable	Promotion premium-cost ratio σ .	Measures whether the career path is low-stakes or high-stakes.
Firm type	Productivity-cost ratio θ .	Determines optimal promotion stakes and optimal bias policy.
Main outcome	Investment gap, promotion gap, firm profit, and optimal bias.	Links individual incentives to firm culture and diversity.
Applications	Analysts, lending, fund flows, banking careers, founders, academia.	Shows how subtle bias changes empirical interpretation.

13.2 Mechanisms

Mechanism	Economic logic	Main prediction
Plausible deniability	Bias is exercised when subjective signals can justify the favored choice.	Individual biased decisions are hard to prove.
Overcompensation	Red invests more to avoid ties where Blue is favored.	Red can have higher observable qualifications in low-stakes contests.
Discouragement	Red expects a lower return to investment because promotion is biased.	Red underinvests in high-stakes contests.
Achievement gap	Investment choices differ before the promotion decision.	Skill gaps can be endogenous responses to bias.
Favoritism gap	Blue is favored conditional on close qualifications.	Promotion gaps can persist even when Red invests more.
Firm polarization	Bias is profitable in low- θ firms but costly in high- θ firms.	Progressive and conservative cultures coexist across firms.

13.3 Subtle versus overt discrimination

Dimension	Subtle discrimination	Overt discrimination
Verifiability	Cannot be objectively established from a single act.	Can be inferred when the chosen candidate is clearly less qualified.
Typical location	Close calls and subjective evaluation margins.	Large objective qualification gaps.
Decision-maker defense	Can appeal to potential, fit, private signals, or subjective quality.	Plausible deniability is weak or absent.
Effect on Red's effort	Can induce overcompensation or discouragement.	More directly discourages the unfavored group.
Outcome tests	May fail because choices are among close substitutes.	More likely to show worse marginal outcomes for favored group.
Policy relevance	Requires attention to subjective discretion and tie-breaking.	Easier to target through formal rules against explicit unequal treatment.

13.4 Main predictions

Prediction	Low-stakes setting	High-stakes setting
Dominant force	Overcompensation.	Discouragement.
Red's investment	Red invests more than Blue.	Red invests less than Blue.
Observable achievement gap	Can favor the discriminated group.	Favors the favored group.
Promotion gap	Can remain positive despite Red's overinvestment.	Increases because investment and favoritism reinforce each other.
Firm incentive to reduce bias	Weak; bias can raise profit.	Strong; bias lowers profit by discouraging valuable investment.
Likely firm culture	Conservative or tolerant of subtle bias.	Progressive or antidiscrimination-oriented.

13.5 Applications and empirical interpretation

Context	Subtle-bias margin	Interpretation
Equity and credit analysts	Accuracy is objective; optimism or client orientation is subjective.	Unfavored analysts may be more accurate because they overinvest in objective skill.
Lending	Loan officers exercise discretion when hard information is inconclusive.	Lower minority approval rates can arise even without safer approved minority borrowers.
Fund flows	Investors break ties among similar funds using manager characteristics.	Discrimination can affect flows without detectable return differences.
Banking careers	Subjective leadership assignments and promotion judgments.	Promotion gaps may reflect both direct favoritism and endogenous effort differences.
Entrepreneurship	Investors evaluate founders across subjective stages.	Expected later-stage bias can change early human capital and startup investments.
Academia	Credit attribution, grant review, and evaluations of potential.	Unfavored groups may overinvest in visible quality, writing, or solo-authored signals.

14 Original figures

14.1 Figure 1 and Figure 2

Read: Figure 1 plots equilibrium investment levels for Red and Blue as the promotion premium-cost ratio rises. Figure 2 plots the resulting promotion gap and decomposes it into achievement and favoritism components.

Economic message: Low-stakes promotions can induce Red to overinvest, while high-stakes promotions lead to discouragement. The promotion gap reflects both endogenous skill choices and biased tie-breaking.

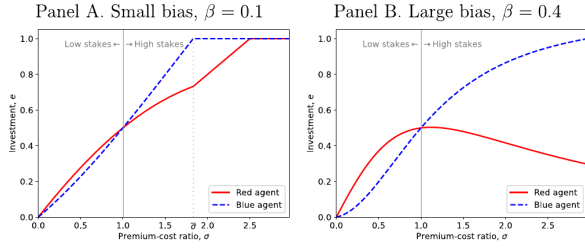


Figure 1. Equilibrium investments. This figure plots the equilibrium investments of Blue and Red, e_b^* and e_r^* , as functions of the premium-cost ratio, σ , for two levels of subtle bias, $\beta_1 = 0.1$ and $\beta_2 = 0.4$. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 1. Equilibrium investments.

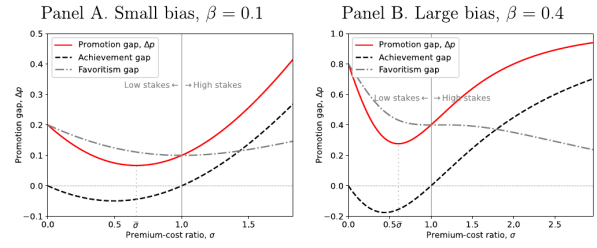


Figure 2. Equilibrium promotion gap. This figure plots the equilibrium promotion gap, Δp^* , as a function of the premium-cost ratio, σ , for two levels of subtle bias, $\beta_1 = 0.1$ and $\beta_2 = 0.4$. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 2. Equilibrium promotion gap.

14.2 Figure 3 and Figure 4

Read: Figure 3 shows how the optimal premium-cost ratio, profit, and promotion gap vary with the productivity-cost ratio for a fixed subtle bias. Figure 4 lets the firm choose the subtle bias and shows the resulting policy discontinuity.

Economic message: Higher-productivity firms choose higher promotion stakes and are hurt more by discouragement. When firms can choose culture or bias, low-productivity firms may choose maximum subtle bias, while high-productivity firms choose no bias.

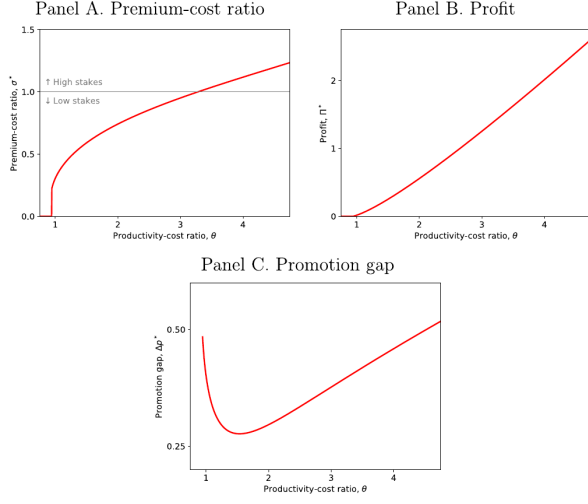


Figure 3. Optimal premium-cost ratio, profit, and promotion gap. This figure plots the optimal premium-cost ratio, σ^* , profit, π^* , and promotion gap, Δp^* , as functions of the productivity-cost ratio, θ , for a given level of subtle bias ($\beta = 0.4$). (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 3. Optimal premium-cost ratio, profit, and promotion gap.

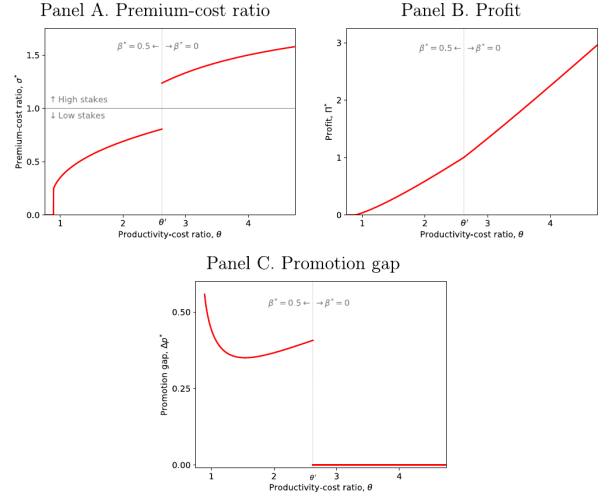


Figure 4. Optimal subtle bias and resulting premium-cost ratio, profit, and promotion gap. This figure plots the optimal premium-cost ratio, σ^* , profit, π^* , and promotion gap, Δp^* , as functions of the productivity-cost ratio, θ . For $\theta < \theta'$, the firm optimally chooses $\beta = 0$, and for $\theta > \theta'$, the firm optimally chooses $\beta = 0.5$. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 4. Optimal subtle bias and resulting outcomes.